

	Course:	Database Systems	Course Code:	CS2005
	Program:	BS (CS)	Semester:	Spring 2026
	Date:	04-Feb-2026	Total Marks:	10
	Quiz:	1 (Relational Model)	Weight:	
	Section:		Max. Time:	10 min.
	Roll No:		**	

Q. Consider the following relational database. The column 'RollNo' in grade table is a foreign key (from student table) and the referential integrity constraint is on Delete/Update cascade. The column 'CCode' in grade table is also a foreign key (from course table) and referential integrity constraint is on Delete/Update no action.

STUDENT PK					GRADE FK FK			COURSE PK		
RollNo	Name	Login	Age	GPA	RollNo	CCode	LGrade	CCode	Title	CrHrs
int	Char(20)	Char(30)	int	Decimal(3,2)	int	Char(6)	Char(2)	Char(6)	Char(20)	int
2200	Tahreem	tahreem@cs	17	3.3	2200	CS2006	A	CS1002	CP	3
2400	Isbah	isbah@cs	16	3.1	2200	CS1002	B	CS2001	DS	3
2600	Izaan	izaan@ee	18	2.6	2800	CS1002	B	CS2005	DB	4
2800	Isbah	isbah@ee	19	3.6	2800	CS2001	C	CS2006	OS	4
3200	Alia	alia@math	18	3.3						

Apply the following operations on the above database. State if the operation would be carried out successfully or not. In case of successful operation indicate the changes that will be made to the above database. Also state all the integrity constraints violated by each operation, if any. Please note that all operations are independent.

- DELETE FROM course WHERE CrHrs=4;
- INSERT INTO grade VALUES (2200, 'CS2004', 'AB');
- UPDATE student SET RollNo=1111, WHERE Name='Isbah';
- DELETE FROM student WHERE age < 18;
- UPDATE grade SET RollNo=3200 WHERE LGrade='B';

1. CrHrs = 4 on two rows with code CS2005 and CS2006, 2005 isn't present as foreign key of Grade, but 2006 is present as Foreign key in Grade Table, but since can't apply an action on CS2006, so it remains same, referential integrity constraint. ✓
2. Can't insert because CS2004 not part of course table's referential integrity constraint. ✓
3. not valid because RollNo is Primary key and it violates primary key unique constraint. ✓
4. In student table with RollNo 2400 will be deleted and also 2200 in grade table RollNo 2200 both row deleted. ✓
5. Valid X (causes duplicate PKs)